Andrew Zhao

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https://zhao-andrew.github.io/

Research Interests

Quantum information theory; many-body physics; quantum algorithms; computational complexity; quantum learning theory

Appointments

Sandia National Laboratories (Livermore, CA)
Senior Member of Technical Staff 2026–present
Gil Herrera Postdoctoral Fellow 2024–2026

Simons Institute for the Theory of Computing (University of California, Berkeley)
Visiting Scientist Starting Mar 2027

Education

PhD in Physics, University of New Mexico 2018–2024
Advisor: Akimasa Miyake
Dissertation: *Learning, Optimizing, and Simulating Fermions with Quantum Computers*
Chairman's Award for Best Dissertation, Department of Physics and Astronomy, 2024

BSc in Physics, University of Maryland, College Park 2013–2018

Research Experience

Google Quantum AI (San Francisco, CA) 2022
Student Researcher (Host: Nicholas Rubin)

Joint Center for Quantum Information and Computer Science (University of Maryland) 2016–2017
Undergraduate Research Assistant (Mentor: Shelby Kimmel)

Funding

- **"Foundations for quantum simulation of warm dense matter"** (Renewal) 2026–2029
  Co-Investigator (PI: Andrew Baczewski). Total award: US$2.1M. Funded by the U.S. Department of Energy, Office of Science, Fusion Energy Sciences Program.
- **"Investigating quantum algorithms and complexity for applications to physical simulation"** 2024–2026
  Principal Investigator. Total award: US$236K. Funded by the Computing & Information Sciences Laboratory Directed R&D Program at Sandia National Laboratories, as part of the Gil Herrera Fellowship.

Publications

* Equal contribution † Alphabetical ordering

Refereed Articles

7. **Andrew Zhao**, "Learning the Structure of Any Hamiltonian from Minimal Assumptions," *Proceedings of the 57th Annual ACM Symposium on Theory of Computing (STOC)*, 1201, 2025. *Also in QIP 2025.*
6. **Andrew Zhao**, Nicholas C. Rubin, "Expanding the reach of quantum optimization with fermionic embeddings," *Quantum* **8**, 1451, 2024.
5. **Andrew Zhao**, Akimasa Miyake, "Group-theoretic error mitigation enabled by classical shadows and symmetries," *npj Quantum Information* **10**, 57, 2024.

4. Ryan Babbush, William J. Huggins, Dominic W. Berry, Shu Fay Ung, **Andrew Zhao**, David R. Reichman, Hartmut Neven, Andrew D. Baczewski, Joonho Lee, "Quantum simulation of exact electron dynamics can be more efficient than classical mean-field methods," *Nature Communications* **14**, 4058, 2023.
3. Austin K. Daniel, Yinyue Zhu, C. Huerta Alderete, Vikas Buchemmavari, Alaina M. Green, Nhung H. Nguyen, Tyler G. Thurtell, **Andrew Zhao**, Norbert M. Linke, Akimasa Miyake, "Quantum computational advantage attested by nonlocal games with the cyclic cluster state," *Physical Review Research* **4**, 033068, 2022.
2. **Andrew Zhao**, Nicholas C. Rubin, Akimasa Miyake, "Fermionic Partial Tomography via Classical Shadows," *Physical Review Letters* **127**, 110504, 2021.
1. **Andrew Zhao**, Andrew Tranter, William M. Kirby, Shu Fay Ung, Akimasa Miyake, Peter J. Love, "Measurement reduction in variational quantum algorithms," *Physical Review A* **101**, 062322, 2020.

Preprints

8. Chaithanya Rayudu, Jun Takahashi, Ojas Parekh, **Andrew Zhao**, "Efficient classical simulation of Fermi-Hubbard models with Kramers symmetry," arXiv:2611.10039, 2026.
7. Tong Jiang,* **Andrew Zhao**,* Shashwat Kumar, Jigyasa Nigam, Eliott Rosenberg, Ryan Babbush, Andrew D. Baczewski, William J. Huggins, Joonho Lee, "Machine learning quantum data to accelerate quantum Monte Carlo," arXiv:2611.08151, 2026.
6. Jiaqing Jiang, Nathan Ju, Ojas Parekh, Chaithanya Rayudu, **Andrew Zhao**,† "Polynomial-time classical and quantum simulation of quantum impurity models," arXiv:2610.02167, 2026.
5. Vishnu Iyer, Bryan O'Gorman, Ojas Parekh, Kevin Thompson, **Andrew Zhao**,† "Optimal testing of fermionic and bosonic Gaussian states," arXiv:2609.39113, 2026.
4. Chaithanya Rayudu, Takahiro Misawa, **Andrew Zhao**, Jun Takahashi, "Lee-Yang theorem for fermions," arXiv:2609.23942, 2026.
3. Pablo Rodenas-Ruiz, **Andrew Zhao**, Joonho Lee, "Quantum Solvers for Nonlinear Matrix Equations in Quantum Chemistry," arXiv:2605.16189, 2026.
2. Aria Christensen, **Andrew Zhao**, "Learning fermionic linear optics with Heisenberg scaling and physical operations," arXiv:2602.05058, 2026.
1. Brent Harrison, Vishnu Iyer, Ojas Parekh, Kevin Thompson, **Andrew Zhao**,† "Fermionic Insights into Measurement-Based Quantum Computation: Circle Graph States Are Not Universal Resources," arXiv:2510.05557, 2025. *In TQC 2026.*

Patents

1. Nicholas C. Rubin, **Andrew Zhao**, "Solving quadratic optimization problems over orthogonal groups using a quantum computer," US Patent US 2024 0296367 A1, 2024.

Talks

"Polynomial-time classical and quantum simulation of quantum impurity models"

- **Invited:** *Hon Hai (Foxconn) Quantum Computing Seminar* — Nov 2026
- **Invited:** *University of Waterloo, Institute for Quantum Computing, Gosset Group Meeting* — Sep 2026

"Quantum solvers for nonlinear matrix equations in quantum chemistry"

- **Invited:** *University of Sherbrooke, Institut quantique Seminar* — Sep 2026
- **Keynote:** *Fields Institute Symposium on Recent Advances on Quantum Algorithms for Differential Equations* — Aug 2026

"Learning fermionic linear optics with Heisenberg scaling and physical operations"

- Contributed: *Machine Learning to Advance Quantum Technologies (ML4QT)* — Sep 2026

"Fermionic insights into MBQC: Circle graph states are not universal resources"
- Contributed: *Southwest Quantum Information and Technology (SQuInT)* Oct 2026
- Contributed: *University of Cambridge, Girton College, Foundations of Quantum Technologies* Sep 2026
- Contributed: *Theory of Quantum Computation, Communication and Cryptography (TQC)* Sep 2026
- **Invited:** *Phasecraft Workshop on Quantum Algorithms* Jun 2026

"Toward scalable auxiliary-field quantum Monte Carlo on quantum computers"
- Contributed: *International Congress on Quantum Chemistry (ICQC)* Jun 2026

"Hamiltonian learning: A survey of recent progress"
- **Invited:** *University of Sherbrooke, Institut quantique Colloquium* Mar 2025

"Learning the structure of any Hamiltonian from minimal assumptions"
- **Invited:** *Visa Research Quantum Seminar* Feb 2026
- Contributed: *ACM Symposium on Theory of Computing (STOC)* Jun 2025
- Contributed: *APS Global Physics Summit* Mar 2025
- Contributed: *Quantum Information Processing (QIP)* Feb 2025
- Contributed: *Southwest Quantum Information and Technology (SQuInT)* Oct 2024

"Quantum relaxation for quadratic programs over orthogonal matrices"
- **Invited:** *INFORMS Annual Meeting* Oct 2023

"Group-theoretic error mitigation enabled by classical shadows and symmetries"
- **Invited:** *University of Oxford, Department of Materials, Benjamin Group Seminar* Nov 2023
- Contributed: *Southwest Quantum Information and Technology (SQuInT)* Oct 2023
- **Invited:** *Sandia Quantum Computer Science Seminar* Jun 2023
- **Invited:** *UC Berkeley Quantum Many-Body Seminar* Jan 2023
- Contributed: *APS March Meeting* Mar 2023

"Fermionic partial tomography via classical shadows"
- Contributed: *APS March Meeting* Mar 2021

"Measurement reduction in variational quantum algorithms"
- Contributed: *APS March Meeting* Mar 2020

Mentorship

Sandia National Laboratories
- Ramya Bhaskar, *PhD student intern (University of Washington)* 2024–2025
- Aria Christensen, *PhD student intern (Ohio State University)* 2024–2026

Teaching

Courses
- General Physics (PHYC 151), Graduate Supplemental Instructor Spring 2019
  University of New Mexico, Department of Physics and Astronomy
- General Physics Laboratory (PHYC 151L), Graduate Lab Instructor Fall 2018
  University of New Mexico, Department of Physics and Astronomy

Guest Lectures
- "Classical shadows are both simpler and more complicated than you think" Mar 2025
  University of Sherbrooke, Institut quantique, seminar
- "Quantum metrology and distributed quantum sensing" Jun 2023
  University of New Mexico, Center for Quantum Information and Control, summer course

- "Random unitary evolution, t-designs, and applications to quantum chaos" Jul 2021
  University of New Mexico, Center for Quantum Information and Control, summer course
- "Continuous symmetries and O(N) models" Jun 2020
  University of New Mexico, Center for Quantum Information and Control, summer course
- "Simple Lie groups and simple Lie algebras" Jun 2019
  University of New Mexico, Center for Quantum Information and Control, summer course

Professional Service

Committees

- QIP Program Committee 2027
- Sandia Gil Herrera Fellowship Selection Committee 2026

Journal Referee

Physical Review Letters, Physical Review A, Physical Review Research, PRX Quantum, Nature Physics, Nature Communications, npj Quantum Information, Quantum, Communications in Mathematical Physics, Journal of Chemical Theory and Computation

Conference Referee

Quantum Information Processing (QIP), Theory of Quantum Computation, Communication and Cryptography (TQC), Quantum Simulation (QSim), Quantum Computing Theory in Practice (QCTiP), Symposium on Discrete Algorithms (SODA)

Grant Proposal Reviewer

Sandia LDRD Program

Outreach

- Summer instructor for BERQ *(Lawrence Berkeley National Laboratory & Berkeley City College)* 2026
- Summer instructor for QCaMP (*Lawrence Berkeley & Sandia National Laboratories)* 2026